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An efficient reliability method combining adaptive Support Vector Machine and Monte Carlo Simulation

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ABSTRACT

To enhance computational efficiency in reliability analysis, metamodeling has been widely adopted for reliability assessment. This work develops an efficient reliability method which takes advantage of the **A**daptive **S**upport **V**ector **M**achine (ASVM) and the **M**onte **C**arlo **S**imulation (MCS). A pool-based ASVM is employed for metamodel construction with the minimum number of training samples, for which a learning function is proposed to sequentially select informative training samples. Then MCS is employed to compute the failure probability based on the SVM classifier obtained. The proposed method is applied to four representative examples, which shows great effectiveness and efficiency of ASVM-MCS, leading to accurate estimation of failure probability with rather low computational cost. ASVM-MCS is a powerful and promising approach for reliability computation, especially for nonlinear and high-dimensional problems.

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1. Introduction

The assessment of failure probability is very important for a structural system whose input parameters are generally subjected to different levels of uncertainty. Therefore, it is interesting to perform probabilistic analysis to account for the uncertainties of input parameters. Considering a structural system whose model response Y is given by a computational model T:

$$Y = T(\boldsymbol{X}) \tag{1}$$

where $\boldsymbol{X} \in \mathbb{R}^d$ is a vector representing input parameters of this physical system and d is the dimension of the problem. Given a set of values of input vector $\boldsymbol{X}$, Eq. (1) may give the corresponding model response. In reality, the uncertainty in the input parameters does exist, due to, for example, inherent soil variability or measurement errors. In order to account for these uncertainties, input parameters are often modelled as random variables following prescribed distributions, like the normal distribution or the lognormal distribution. The model response Y is therefore also a random variable.

Under this circumstance, the failure probability of a structural system can be defined by the following multi-dimensional integral expression:

$$P_f = \int \cdots \int I[G(\boldsymbol{x})] f(\boldsymbol{x}) d\boldsymbol{x} \tag{2}$$

where $\boldsymbol{x}$ is a realization of the d-dimensional random vector $\boldsymbol{X}$ sampled from the joint probability density function (PDF) $f(\boldsymbol{x})$; $G(\boldsymbol{x}) = 0$ is the limit state surface of the structural system, $G(\boldsymbol{x}) < 0$ defining the failure domain and $G(\boldsymbol{x}) > 0$ the safe domain. The indicator function $I[G(\boldsymbol{x})]$ is equal to 1 for $G(\boldsymbol{x}) \leqslant 0$, otherwise $I[G(\boldsymbol{x})] = 0$. The limit state surface G is often determined by the computational model T and a given threshold. For instance, if the computational model T predicts the deformation of a specified position of a considered system, and the maximum allowable deformation at that position is denoted by S', then the limit state function $G(x)$ can be written as:

$$G(\boldsymbol{x}) = T(\boldsymbol{x}) - S' \tag{3}$$

It is common to transform the random variables $\boldsymbol{x}$ into a standard space, which is composed of independent standard normal variables $\boldsymbol{u}$, by means of the Nataf transformation [40] or the Rosenblatt transformation [41]. Then the failure probability can be reformulated in the standard normal space as:

$$P_f = \int \cdots \int I[g(\boldsymbol{u})] \varphi(\boldsymbol{u}) d\boldsymbol{u} \tag{4}$$

where $\varphi(\boldsymbol{u})$ is the standard normal PDF and $g(\boldsymbol{u}) = G(\boldsymbol{x}(\boldsymbol{u}))$ represents the transformed limit state function in the standard normal space.

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The exact calculation of the integral of Eq. (4) is generally not practically feasible, especially for high-dimensional problems. Alternatively, some approximated approaches have been developed to address this problem, such as the first-order and second-order reliability method (FORM/SORM), the Monte Carlo Simulation (MCS), the response surface method (RSM).

The most straightforward and robust one is the MCS, and it proceeds in three steps: (1) randomly sampling n_{MC} sets of input parameters according to underlying PDFs, (2) repeatedly running G for all samples and (3) post-calculating the failure probability or the statistical moments. Then, Eq. (4) can be approximated by:

$$\widehat{P}_f = \frac{1}{n_{MC}} \sum_{i=1}^{n_{MC}} I[G(\boldsymbol{u}_i)] \tag{5}$$

where $\boldsymbol{u}_i$ represents the ith MC samples. Eq. (5) is an unbiased estimate of failure probability and its coefficient of variation (COV) is evaluated by:

$$COV(\widehat{P}_f) = \sqrt{\frac{1-\widehat{P}_f}{n_{MC}\widehat{P}_f}} \tag{6}$$

The classical Monte Carlo Simulation is dimension-independent, and often regarded as a standard reference for the test of other probabilistic methods because of its versatility and robustness. However, it suffers from a quite low computational efficiency. This makes it hardly applicable for computationally expensive models, such as the finite element or finite difference numerical solvers, especially when the failure probability is rather small.

Recently, a metamodel-based Monte Carlo method has gained considerable attention in the community of reliability analysis. The core idea of this method is to firstly build an analytical metamodel (or the response surface) with limited calls to the original model based on the Design of Experiments (DoE), and then perform Monte Carlo Simulation on the obtained metamodel. The metamodel should be capable of catching the global behavior of the original model responses. There are several mathematical tools available to construct such a metamodel, e.g. polynomial chaos expansion (PCE), Kriging model and Artificial Neural Networks (ANN). The polynomial chaos expansion (PCE) is used to build a metamodel. A Collocation-based Stochastic Response Surface Method (CSRSM) on the basis of PCE was applied by researchers to perform probabilistic analysis [1,2]. However, the PCE suffers from the "curse of dimensionality". The size of the DoE rapidly increases with the number of the input variables and with the PCE order. In order to enhance the applicability of PCE, a sparse polynomial chaos expansion (SPCE) was proposed by Blatman and Sudret [3,4] which results in less terms than a full PCE. The Kriging model [5,6] and ANN [7,8] also attract more and more interest for probabilistic assessment.

Another powerful vehicle for the metamodel construction is the Support Vector Machine (SVM) developed by Vapnik [9,10] in the field of statistical learning theory. It has the following merits [11–15]: (1) In a SVM decision function, only support vectors which are a small fraction of the training set contribute to the model predictions (see Eq. (10)). This feature makes the model predictions more efficient compared with other surrogate models, the Kriging model for example. (2) The SVM makes advantage of the Structural Risk Minimization that minimizes an upper bound on the generalization error. This also gives SVM good performance on avoiding overfitting and on generalization. (3) The SVM deals with classifications of model responses, for example the safe state (labeled as +1) and failed state (labeled as -1) in structural stability, which makes it applicable for Monte Carlo Simulation for which only the "sign" of the model response instead of the exact value is of interest in the computation of failure probability. (4) The SVM is able to bypass the curse of dimensionality and handle highly nonlinear problems effectively, e.g. non-convex and disjoint limit state functions.

Many papers have been devoted to application of the Support Vector Machine (SVM) to reliability analysis recently. The pioneering work appears to be done by Rocco and Moreno [11] who firstly combined Monte Carlo Simulation and Support Vector Machine for reliability analysis. Hurtado, J. E. [16] developed a reliability algorithm based on Importance Sampling and SVM which is shown to be rather computationally efficient with respect to conventional Importance Sampling method. Li et al. [15], Zhao [17] and Zhao et al. [18] proposed a reliability method combining the SVM and the FORM and tested their methods on slope and tunnel reliability analysis. Tan et al. [19] reported four response surface methods which are respectively based on radial basis neural network (RBFN) and SVM for reliability analysis. No obvious difference was found between RBFN-based RSMs and SVM-based RSMs. Bourinet et al. [20] proposed an approach termed as "^{2}SMART" which combines Subset Simulation and SVM (SS-SVM). This approach is able to deal with reliability problems involving small failure probabilities and large numbers of random variables (up to a few hundreds). Ji et al. [21] applied the least-squares support vector (LS-SVM) in combination of Monte Carlo Simulation to assess slope system reliability. It is shown that the LS-SVM is effective to evaluate system probability of a complex slope involving several failure regions. The aforementioned research on application of SVM to reliability analysis mainly follows the procedures of firstly constructing the response surface using SVM based on the Design of Experiments (DoE) and then computing the failure probability or reliability index by using probabilistic approaches (e.g. FORM or MCS) thanks to the surrogate model obtained. Therefore, the effectiveness and efficiency of these methods highly depends on how to train the SVM model, with a required accuracy at the cost of the smallest size of DoE (training samples). The uniform design for the DoE is highlighted by Li et al. [15] and Ji et al. [21] for this purpose. Another more efficient technique is to use an adaptive SVM which aims to build a surrogate by sequentially selecting informative samples. Several adaptive SVM models can be found in Song et al. [14], Tong and Koller [22], Basudhar and Missoum [23].

This work aims to develop an efficient reliability method named ASVM-MCS which combines **A**daptive **S**upport **V**ector **M**achine in conjunction with **M**onte **C**arlo **S**imulation. The remainder of this paper is outlined as follows. The basic theory about how a SVM classifier works is presented in the Section 2. The following section introduces adaptive SVM and details the proposed ASVM-MCS algorithm. Four representative examples for the validation of the ASVM-MCS are given in Section 4. This paper ends up with a conclusion at Section 5.

2. A basic introduction to Support Vector Machines

The support vector machine (SVM) is an efficient classifier originated from pattern recognition in machine learning. For a two-category problem, the SVM training algorithm aims at building a hyperplane that separates all training data of one category from those of the other category. The SVM has been successfully applied to wide applications [22,24], ranging from pattern recognition, hand-written characters recognition, text classification to biological sciences. Recently, it has attracted more and more attention in reliability analysis. This section simply presents an overview of the SVM algorithm. For more details, the reader is referred to Vapnik [9,10] .

2.1. Linear decision function

Given a two-class problem, suppose we have N sets of labeled training datum $\{\boldsymbol{x}_i, y_i\}$, where vector $x_i \in \mathbb{R}^d$, $y_i \in \{-1, +1\}$, as sketched in Fig. 1. SVM aims to search for an optimal hyperplane which is a separating function (or a decision function) for which all vectors labeled as -1 are located on one side and all vectors labeled as +1 on the other side. The optimal hyperplane is the one that has the largest distance to the nearest training samples of any class (maximum margin). Considering a possible hyperplane (the SVM decision function):

$$f(\boldsymbol{x}) = \boldsymbol{w}^T\boldsymbol{x} + b = 0 \tag{7}$$

where the vector $\boldsymbol{w}$ is perpendicular to the hyperplane and b is a scalar parameter. All training points should satisfy the following constraints:

$$y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) - 1 \geqslant 0 \quad i = 1, \ldots, N \tag{8}$$

This constraint ensures that there is no sample existing inside the margin. The margin is calculated by $2/\|\boldsymbol{w}\|$. Therefore, determining the optimal hyperplane with maximum margin is equivalently reduced to the following optimization problem:

$$\min_{\boldsymbol{w},b} \left\{ \frac{1}{2}\|\boldsymbol{w}\|^2 \right\} \tag{9}$$

$$s.t. \quad y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) - 1 \geqslant 0, \quad i = 1, \ldots, N$$

Introducing Lagrange multipliers, the above optimization problem is transformed as a convex quadratic optimization problem whose analytical solution is available. Please note that, as seen in Fig. 1, only those closest points have nonzero Lagrange multipliers. Such points that actually contribute to the SVM-building are termed as support vectors.

The classification of any arbitrary point $\boldsymbol{x}$ to be predicted is determined by the following function:

$$c(\boldsymbol{x}) = sign\left[\sum_{i=1}^{N_{SV}} \alpha_i y_i \boldsymbol{x}_i^T \boldsymbol{x} + b \right] \tag{10}$$

where N_{SV} is the number of support vectors, which is generally a small fraction of the total number of training samples.

In order to extend SVM to cases where it is impossible to separate all data points, meaning the optimization problem is not solvable, a penalty parameter C is introduced to allow for a soft margin. The soft margin parameter C permits the misclassification and should be specified by the user. It should note that C is the upper bound of α_i's, and increasing C generates a stricter separation between classes. If we reduce C towards 0, it makes misclassification less important; if we increase C to infinity, it means no misclassification is allowed.

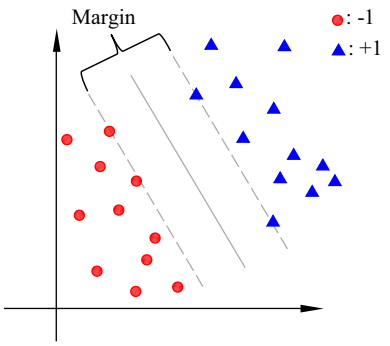


Fig. 1. A schematic for a linear SVM classifier (red circles representing class −1 and blue triangles class +1).

2.2. Nonlinear decision function

This section presents how SVM works in the case of nonlinearly separable samples. The kernel trick is used to address this problem. The main idea is to map the input data into a higher-dimensional feature space in which the classification becomes possible. Fig. 2 shows an illustration of such a transformation. In the feature space, the non-linear classification problem becomes linear, and the formulation is analogous to the linearly separable case except for replacing $\boldsymbol{x}_i^T\boldsymbol{x}$ by $\Psi(\boldsymbol{x}, \boldsymbol{x}_i)$. The prediction of classification of a point $\boldsymbol{x}$ is then expressed as:

$$c(\boldsymbol{x}) = sign\left[\sum_{i=1}^{N_{SV}} \alpha_i y_i \Psi(\boldsymbol{x}, \boldsymbol{x}_i) + b \right] \tag{11}$$

where $\Psi(\boldsymbol{x}, \boldsymbol{x}_i)$ is a Kernel function: linear kernel $\Psi(\boldsymbol{x}, \boldsymbol{x}_i) = \boldsymbol{x}_i^T\boldsymbol{x}$, polynomial kernel $\Psi(\boldsymbol{x}, \boldsymbol{x}_i) = (\boldsymbol{x}_i^T\boldsymbol{x} + 1)^p$, radial basis (or Gaussian) kernel $\Psi(\boldsymbol{x}, \boldsymbol{x}_i) = \exp(-\|\boldsymbol{x} - \boldsymbol{x}_i\|^2/2\sigma^2)$, p and σ being constants. The linear kernel is used for a linear classification, while for a non-linear classification problem, we should choose the polynomial kernel or Gaussian kernel. In addition to soft margin parameter C, the kernel's parameters should be selected for a specified kernel function. The effectiveness of a SVM highly depends on the selection of kernel's parameters.

3. The proposed method: ASVM-MC

This section presents the adaptive Support Vector Machine combined with the Monte Carlo simulation. The algorithm consists of generating a Monte Carlo population based on which the Support Vector Machine is trained. Instead of randomly selecting training samples from the Monte Carlo population, active learning technique aims to choose targeted training data that is most informative. The first section briefly introduces the adaptive Support Vector Machine and the second part presents the specified algorithm for ASVM-MC.

3.1. Adaptive SVM

The primary difference between a passive learner and an active learner lies in how they enrich the Design of Experiments (or the training set). A passive learner enriches it randomly; on contrary, an active learner chooses the next training point on purpose with the help of a learning function. An active learner outperforms the passive learner in terms of efficiency. An active learner is able to build a metamodel at the expense of as few training samples as possible. This feature is significant for the reliability analysis with a time-consuming model.

The main idea of an adaptive SVM is to select a training sample which is most informative. Such a sample has the highest probability to be misclassified and is about to change the classifier when it is added to the training group. In the previous work, several approaches have been developed to adaptively update the SVM classifier. Tong and Koller [22] proposed a pool-based adaptive SVM for text classification. The learning function has access to a pool of unlabeled samples and queries the next point that separates a so-called version space into two equal parts as much as possible. Two schemes were developed to find such a sample: (1) Simple Margin, to select the sample that comes closest to the hyperplane in the feature space; (2) MaxMin Margin or Ratio Margin, to check the change of margin size if a candidate sample is labeled as "−1" and if it is labeled as "+1". The second scheme is more stable but it is computationally intensive since multi-training SVMs are involved.

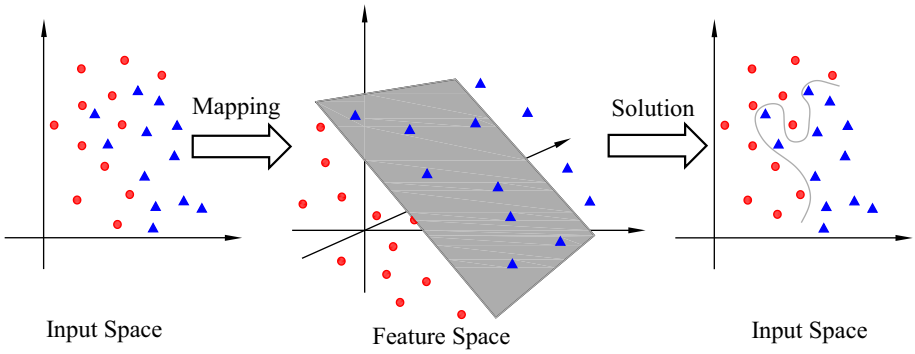


Fig. 2. A schematic for a nonlinear SVM classifier and the Kernel trick.

Basudhar and Missoum [23] proposed an adaptive SVM for probabilistic design and optimization. It states that a new training sample to be selected should be located on the decision boundary and have a given minimum distance from existing training samples. The updating of the classifier proceeds to repeatedly find three kinds of samples on the decision boundary: (1) sample with a given distance from the nearest existing training sample, (2) sample with the maximum distance from the previously added one and (3) sample with maximum minimum distance from existing training samples. Song et al. [14] reported a sequential sampling scheme which is similar to that developed by Basudhar and Missoum [23], but it only requires that the newly added training sample is located within the margin and has the maximum distance from the closest existing one. In the updating algorithms proposed by Basudhar and Missoum [23] and Song et al. [14], a new sample is searched by solving an optimization problem using a genetic algorithm or an active-set algorithm. A possible defect is that these two updating algorithms are highly depended on the global optimization algorithm whose effectiveness and efficiency are normally related to the size of the training set and the dimension of the problem.

3.2. A pool-based active learner: ASVM-MCS

The SVM technique is used to construct a metamodel for reliability analysis. The combination of SVM and Monte Carlo Simulation seems to be firstly implemented by Rocco and Moreno [11]. In this previous work, the training samples for the construction of SVM are selected passively. The presented approach combines the adaptive SVM with the Monte Carlo Simulation. Fig. 3 shows a flowchart for the ASVM-MCS algorithm, and it is based on the following steps:

1. Generate n_{MC} Monte Carlo samples using Latin Hypercube sampling. This Monte Carlo population is termed by U. A metamodel to be constructed should provide good predictions over the whole domain under required accuracy. The uniform sampling strategy helps to achieve this goal. There are several available uniform sampling techniques, such as the Latin Hypercube Sampling (LHS), Uniform Design [25] and LCVT [26]. The LHS is used to generate n_{MC} Monte Carlo samples in the standard normal space in this work. There is a possibility that the size of U is required to increase when the coefficient of variation of failure probability does not satisfy the targeted accuracy (see in Step 7). Under this context, the Nested Latin Hypercube sampling technique (NLHS) [3] can be used to generate new samples in order to maintain the uniformity of U. Please note that U is a pool composed of unlabelled samples.

2. Define the initial Design of Experiments (DoE). The initial DoE should contain both classes of samples. It is preferred to make initial DoE with maximum possible uniformity, on the purpose of catching the global behaviour of the model response. It is suggested to employ the three-Sigma rule to determine the sampling domain of the initial DoE. This uniformity of initial DoE can also be achieved by using the LHS technique. The suitable size of initial DoE (initial training samples), depending on the complexity of the problem, is set to be between 10 and 20 in the examples presented in this work. As the iteration proceeds, the initial DoE is enriched by adding one training sample step by step according to the active learning scheme. The group of training samples (including initial DoE and newly added training samples) is named by T. The enrichment of T will not terminate until the stopping criterion in Step 5 is reached.

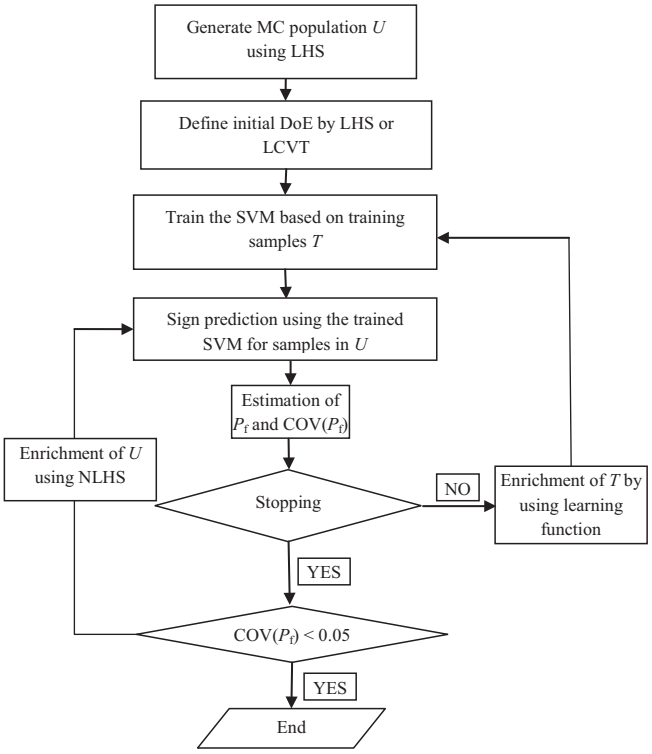


Fig. 3. Flowchart for the ASVM-MCS algorithm.

3. Train the SVM classifier based on training samples. Evaluate $G(\boldsymbol{X})$ for training samples in T and train a SVM classifier using the Statistics and Machine Learning Toolbox in Matlab. The kernel function should be selected for the non-linear classification. Even though some authors concluded that the SVM is kernel-insensitive [11,15], the choice of a kernel function is of primarily importance in machine learning. For example, the Gaussian kernel works well for smooth limit state functions, but it may not give

good results if this assumption is not met. Due to the excellent applicability as well as high flexibility, as many previously published articles did, this work chooses the Gaussian kernel which has a single parameter σ to be tuned. The magnitude of the kernel parameter σ is related to the problem dimension and the complexity of the limit state function to be simulated. The selection of an optimum σ is an ongoing research topic. In this work, the value of σ is determined by using a 5-fold cross validation [20,27]. Besides, the margin parameter is set to be infinite to avoid misclassification in the presented examples. Even though this hard margin setting gives good results in all examples addressed in this work, it should be noticed that it may easily cause overfitting for more complex limit state surfaces.

4. Estimation of the failure probability and its coefficient of variation. The obtained SVM classifier at Stage 3 is used to predict the model response for each sample in U. Then the failure probability P_f is calculated according to the sign of SVM predictions by using Eq. (5), and the coefficient of variation of P_f is also computed by using Eq. (6).

5. Stopping criterion. A stopping criterion is required to stop the updating. Even though the SVM margin should not contain any sample in T (because the margin parameter C is set to be infinite), there exists samples that belong to U. The active learner selects one point located within the margin (from U) and adds it to the training set T at each iteration step. The number of samples that are located in the margin varies as the algorithm proceeds. The ratio of the number of samples located inside the margin to the total number of samples in U can be taken to measure the convergence of the learning process and it is calculated by:

$$\delta_k = \frac{\sum_{i=1}^{n_{MC}} J(\boldsymbol{x}_i)}{n_{MC}} \tag{12}$$

where k is the current iteration number. The indicator function $J(\boldsymbol{x}_i)$ is equal to 1 if $\boldsymbol{x}_i$ is located within the margin, otherwise $J(\boldsymbol{x}_i) = 0$. In order to implement a more stable stopping criterion, the data points of ratio δ_k is fitted by an exponential curve [23]:

$$\widehat{\delta}_k = Ae^{Bk} \tag{13}$$

where $\widehat{\delta}_k$ is the fitted value of δ_k; A and B are parameters of the fitted exponential curve.

Both δ_k and $\widehat{\delta}_k$ should be smaller than a target value ε_1 for a robust convergence. In addition, the absolute value of the slope of the fitted curve is checked and it also should be less than a given value ε_2. Therefore, the stopping criterion in this work is defined as:

$$\begin{cases} \max(\widehat{\delta}_k, \delta_k) < \varepsilon_1 \\ \quad AND \\ 0 < -BAe^{Bk} < \varepsilon_2 \end{cases} \tag{14}$$

6. Actively select the optimal next sample to update the SVM classifier. If the stopping criterion is not met, it means that a SVM classifier trained with less samples, e.g. based on the initial DoE, should be further improved in an attempt to give a good approximate to the original limit state function. This stage is called as the active learning process by which a learning function is designed to identify the next targeted sample in U. Such a candidate sample should simultaneously satisfy two conditions: (1) located closest to the SVM boundary, and (2) in order to avoid redundant information, not be close to existing training samples.

The point which is closest to the SVM classifier has the highest probability to be misclassified, thus it should be taken into account for the updating of the SVM classifier. The distance of a point $\boldsymbol{x}$ to the SVM classifier is denoted by $s(\boldsymbol{x})$ and it is computed by the following expression:

$$s(\boldsymbol{x}) = \frac{\left|\sum_{i=1}^{N_{SV}} \alpha_i y_i \Psi(\boldsymbol{x}, \boldsymbol{x}_i) + b\right|}{||\mathbf{w}||} \tag{15}$$

The new training sample should have a maximum distance from its nearest existing training sample. This condition makes best use of additional information from newly added point. The distance of a point $\boldsymbol{x}$ to its nearest existing training sample is calculated by:

$$d(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{x}_{nearest}\| \tag{16}$$

A learning function $L(\boldsymbol{x})$ which considers a balance between both two conditions is defined by:

$$L(\boldsymbol{x}) = \min_{\boldsymbol{x}} \frac{\bar{s}(\boldsymbol{x})}{\bar{d}(\boldsymbol{x})} \tag{17}$$

$$s.t. \begin{cases} \boldsymbol{x} \in U, \boldsymbol{x}_{nearest} \in T \\ \left|\sum_{i=1}^{N_{SV}} \alpha_i y_i \Psi(\boldsymbol{x}, \boldsymbol{x}_i) + b\right| < 1 \end{cases}$$

where $\bar{s}(\boldsymbol{x})$ and $\bar{d}(\boldsymbol{x})$ are respectively normalised $s(\boldsymbol{x})$ and $d(\boldsymbol{x})$ by their maximum values, $\bar{s}(\boldsymbol{x}) = \frac{s(\boldsymbol{x})}{\max[s(\boldsymbol{x})]}$, $\bar{d}(\boldsymbol{x}) = \frac{d(\boldsymbol{x})}{\max[d(\boldsymbol{x})]}$. Eq. (17) is used to identity the best next point for updating the obtained SVM classifier. This newly selected sample is added into T and the algorithm goes back to Step 3.

7. Enrichment of the Monte Carlo population using NLHS. If the stopping criterion at Stage 5 is satisfied, it is considered that the trained SVM classifier is accurate enough to predict the sign of samples in U. Then the coefficient of variation of P_f is checked. If the coefficient of variation of P_f is higher than 5%, the size of U should be enlarged. In order to keep the uniformity of U, the NLHS technique is used to generate new points to be added into U. The algorithm returns to Step 4. Please note that no previously obtained information is lost.

8. End of the ASVM-MCS algorithm. If the coefficient of variation of P_f is lower than 5%, it indicates that the current size of U is large enough to give an acceptable estimation of the failure probability. The ASVM-MC algorithm terminates, and the failure probability at the last iteration is taken as the final result.

The proposed ASVM-MCS is a kind of pool-based active learning algorithm. A similar algorithm termed as AK-MCS was also proposed by Echard et al. [6] who adopted the active learning Kriging technique in combination of MCS. The difference between ASVM-MCS and AK-MCS lies in several aspects: (1) the uniformity of Monte Carlo population and initial DoE is emphasized in ASVM-MCS by using NLHS technique; (2) ASVM-MCS makes use of the classification feature of SVM which is more efficient in model predictions in comparison to the Kriging model; (3) the active learning strategy and the stopping criterion are completely different between the two algorithms.

4. Examples of application

The performance of the proposed ASVM-MCS is tested on four examples. The former three representative examples have explicit performance function involving complex limit state functions (nonlinearity) and high-dimensional problems (40–250 variables). The last example is a practical problem considering a tunnel face stability in which the performance function is implicit. The proposed ASVM-MCS is compared with other reliability methods in literature. In each example, the failure probability given by the direct Monte Carlo Simulation is taken as a reference value, and the efficiency and accuracy between different approaches are mainly compared in terms of the total number of calls to the limit

state function N_c, and the error ε_{Pf} between the estimated failure probability $\widehat{P}_f$ and the reference value.

4.1. A series system with multiple design points

The first example is a two dimensional series system with four branches. This example was studied by Echard et al. [6], Bourinet et al. [20], Schueremans and Van Gemert [28]. The limit state function is given by:

$$G(x_1,x_2) = min\begin{Bmatrix} 3.0+0.1(x_1-x_2)^2-(x_1+x_2)/\sqrt{2} \\ 3.0+0.1(x_1-x_2)^2+(x_1+x_2)/\sqrt{2} \\ (x_1-x_2)+k/\sqrt{2} \\ -(x_1-x_2)+k/\sqrt{2} \end{Bmatrix}$$

where two independent random variables x_1 and x_2 follow normal distribution, and k is respectively equal to 6 and 7.

The results are presented in Table 1. The failure probability provided by the proposed ASVM-MCS are compared to those given by AK-MCS [6], by SS-SVM [20], by Directional Sampling + Neural Network (DS-NN) and Importance Sampling + Neural Network (IS-NN) [28]. The value of ε_1 and ε_1 in ASVM-MCS for the stopping criterion are respectively set to 4.0×10^{-4} and 1.0×10^{-5} and the size of U is equal to 1.0×10^6.

It is found that the probability of failure estimated by ASVM-MCS, AK-MCS and SS-SVM are very close to that given by direct MCS in both cases, the maximum error being 3.7% which comes from AK-MCS at $k = 7$. These three approaches give better estimate of failure probability than DS-NN and IS-NN; a maximum error of 53.48% is reached in DS-NN in the case of $k = 7$. With respect to the number of calls to the limit state function, ASVM-MCS needs less number of calls than AK-MCS does, for example ASVM-MCS and AK-MCS respectively needs 99 and 126 calls for $k = 6$, 89 and 96 calls for $k = 7$. Compared with SS-SVM, the number of N_c required by SS-SVM for equivalent accuracy is over ten times to that of ASVM-MCS. It is concluded that ASVM-MCS highly outperforms SS-SVM in terms of efficiency, and is comparable to AK-MCS.

Fig. 4 shows a classification boundary provided by ASVM-MCS and the corresponding training samples, while Fig. 5 plots the SVM classifier with the corresponding Monte Carlo samples in U. It is seen that most selected training samples are situated around the real boundary. Even though the SVM classifier does not overlap quite well with the original limit state function at the region close to four sharp corners where Monte Carlo samples are rare, it approximates the limit state function well and makes correct classifications over the domain in which Monte Carlo samples are dense (see in Fig. 5). This feature highly correlates with the learning scheme adopted in this work for which the next training point comes from the Monte Carlo population, and guarantees a good performance of ASVM-MCS.

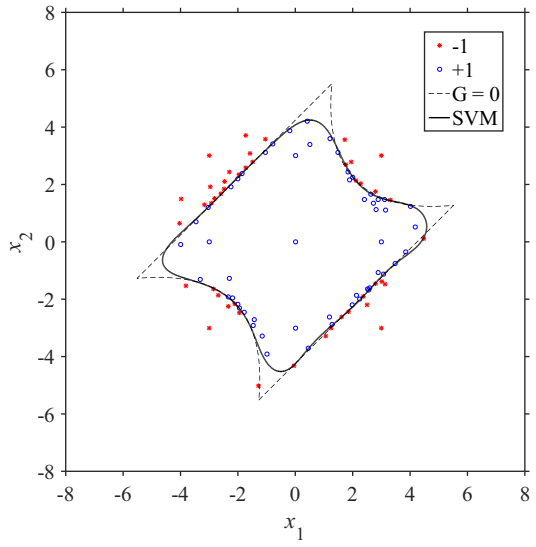


Fig. 4. SVM classifier and its training samples in Example 1 for k = 6.

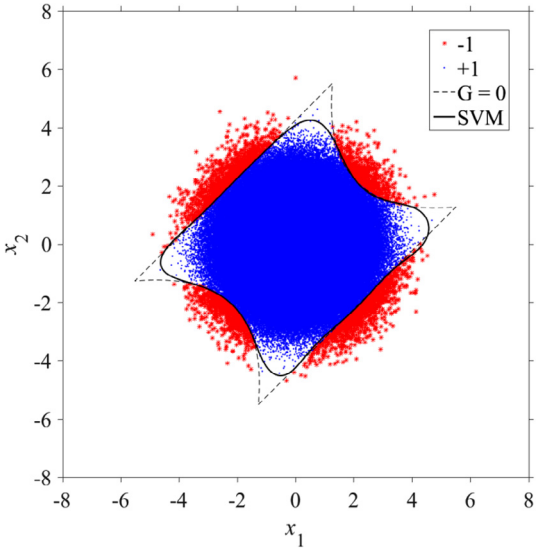


Fig. 5. SVM classifier and the MC population U in Example 1 for k = 6.

The convergence of the ASVM-MCS algorithm is plotted in Fig. 6, which shows percentage of Monte Carlo samples located inside margin δ_k as a function of the number of calls to the limit state function. The black solid line represents the real values of

Table 1
Results of Example 1.

k	Method	N_c	$\widehat{P}_f$	COV($\widehat{P}_f$) (%)	ε_{Pf} (%)
6	Direct MCS	10^6	4.40×10^{-3}	1.7%	–
	ASVM-MCS	99	4.46×10^{-3}	1.5%	1.36
	AK-MCS	126	4.42×10^{-3}	–	0.45
	DS-NN	165	4.1×10^{-3}	–	6.81
	IS-NN	52	5.7×10^{-3}	–	29.54
7	Direct MCS	10^6	2.15×10^{-3}	2.2%	–
	ASVM-MCS	89	2.13×10^{-3}	2.2%	0.93
	AK-MCS	96	2.23×10^{-3}	–	3.72
	SS-SVM	1035	2.21×10^{-3}	1.7%	2.79
	DS-NN	67	1.0×10^{-3}	–	53.48
	IS-NN	125	2.9×10^{-3}	–	34.88

δ_k while the red dashed line is the fitted exponential curve. As expected, the value of δ_k declines as the iteration proceeds.

Fig. 7 presents the evolution of normalised failure probability with respect to the number of calls to the limit state function. The estimated failure probability is normalised by the reference value provided by direction Monte Carlo Simulation. This curve also reflects the convergence of the ASVM-MCS algorithm. It is seen that the normalised $\widehat{P}_f$ starts to converge at around 60 calls, which is earlier than those (99 for $k = 6$ and 89 for $k = 7$) required by the stopping criterion. The remaining calls are mainly required to satisfy the stopping criterion.

This indicates that the stopping criterion functions effectively to give reliable estimate. This probably gives an idea to further reduce the number of calls to the limit state function by checking the convergence of normalised $\hat{P}_f$. But we still choose to use the current stopping criterion in the remaining three examples in order to show the robustness of the proposed ASVM-MCS.

4.2. Dynamic response of a non-linear oscillator

This example deals with a dynamic response of a non-linear undamped system with one degree-of-freedom, as sketched in Fig. 8. This problem has a moderate number of random variables and is investigated by using Directional Sampling + Neural Network (DS-NN) and Importance Sampling + Neural Network (IS-NN) in Schueremans and Van Gemert [28], by using AK-MCS in Echard et al.[6] and by a Complete Quadratic Response Surface with ReSampling (CQ2RS) in Gayton et al. [42]. The limit state function reads:

$$G(c_1, c_2, m, r, t_1, F_1) = 3r - |s_{max}|$$

$$s_{max} = \begin{cases} \frac{2F_1}{m\omega_0^2} & t_1 \geqslant \frac{\pi}{\omega_0} \\ \frac{2F_1}{m\omega_0^2} \sin\left(\frac{\omega_0 t_1}{2}\right) & t_1 < \frac{\pi}{\omega_0} \end{cases}$$

where $\omega_0 = \sqrt{\frac{c_1+c_2}{m}}$, s_{max} is the maximum displacement of the oscillator and r is the yielding displacement of the springs. This system contains six independent random variables, which are normally distributed, and their means and standard deviations are given in Table 2.

Obtained results are summarized in Table 3. The reference value estimated by direct MCS with a number of 1.0×10^6 samples is

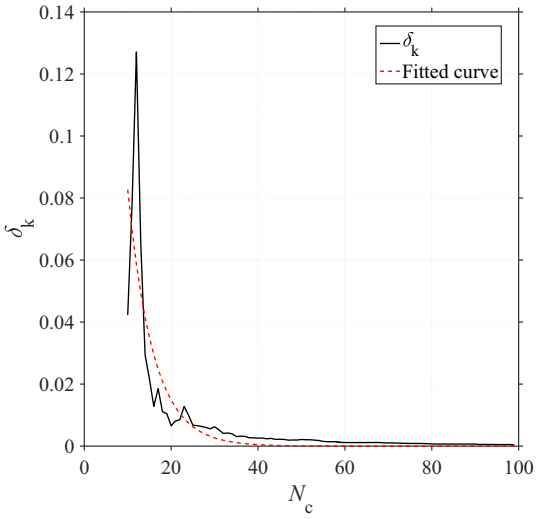


Fig. 6. Convergence curve of ASVM-MCS in Example 1.

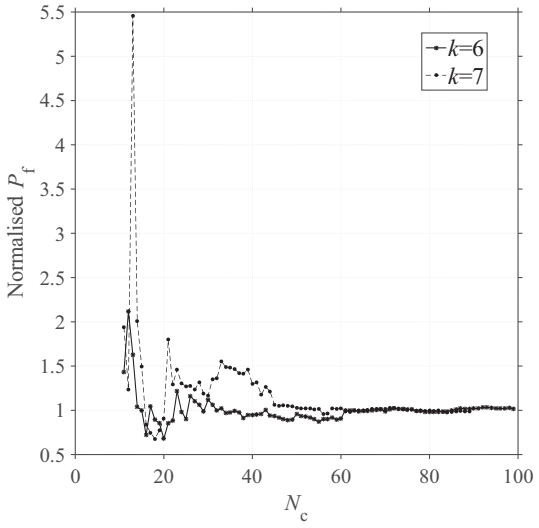


Fig. 7. Evolution of normalised $\widehat{P}_f$ with N_c in Example 1.

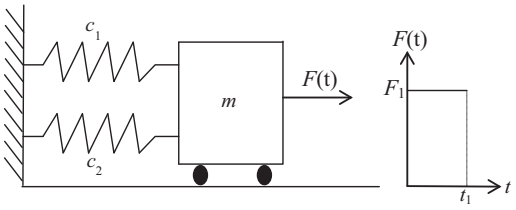


Fig. 8. Sketch of a non-linear oscillator.

Table 2
Statistical properties of random variables in Example 2.

Variables	Mean	Standard deviation	Distribution Type
m	1.0	0.05	Normal
c_1	1.0	0.1	Normal
c_2	0.1	0.01	Normal
r	0.5	0.05	Normal
F_1	1.0	0.2	Normal
t_1	1.0	0.2	Normal

2.87×10^{-2}. The value of ε_1 and ε_1 for the stopping criterion are respectively set to 1.0×10^{-4} and 1.0×10^{-5} and the size of U is equal to 1.0×10^6.

It is found that failure probabilities obtained by ASVM-MCS, by AK-MCS and by DS-NN are quite similar to that estimated by direct MCS. The ASVM-MCS needs only 56 calls to the limit state function, which is the most efficient one, whereas the DS-NN requires the most calls (86 calls). On the contrary, IS-NN provides slightly poor solution, which has the maximum error of 8.01%, though it requires 68 call to the limit state function.

Fig. 9 shows the evolution of normalised failure probability with respect to the number of calls to the limit state function in this example. Again, it is seen that the failure probability converges before the algorithm stops, with only 41 calls to the limit state function.

4.3. High-dimensional example

High dimension still remains to be a challenging topic in reliability analysis due to the "curse of dimensionality". This example aims to apply the proposed ASVM-MCS to a high-dimensional

Table 3
Results of Example 2.

Method	N_c	$\widehat{P}_f$	COV($\widehat{P}_f$) (%)	ε_{Pf} (%)
Direct MCS	1.0×10^6	2.87×10^{-2}	0.6	–
ASVM-MCS	56	2.79×10^{-2}	0.6	2.78%
AK-MCS	58	2.83×10^{-2}	–	1.39%
DS-NN	86	2.8×10^{-2}	–	2.44%
IS-NN	68	3.1×10^{-2}	–	8.01%

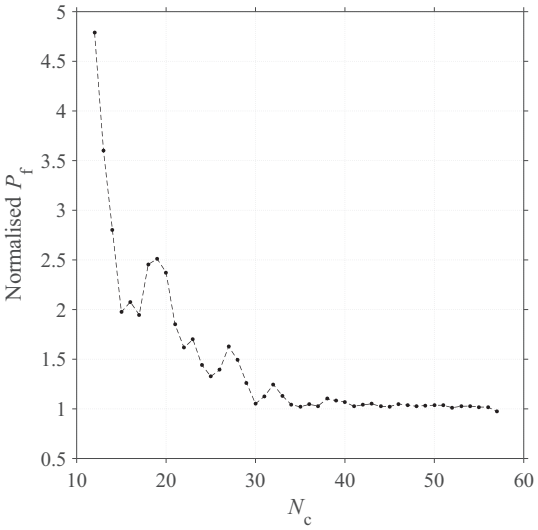


Fig. 9. Evolution of normalised $\widehat{P}_f$ with N_c in Example 2.

problem which was proposed in Rackwitz [29]. Its analytical limit state function reads:

$$G(x_1, \ldots, x_n) = n + 3\sigma\sqrt{n} - \sum_{i=1}^{n} x_i$$

where x_i, $i = 1, \ldots, n$, are independent lognormal random variables with unit means and standard deviations $\sigma = 0.2$; n denotes the number of random variables. This analytical function has a feature that its limit state surface has equal curvatures and its concavity is pointed toward the origin. This means that varying the number of variables hardly alters the shape of the limit state surface as well as the failure probability, $P_f \to 1.35 \times 10^{-3}$ for $n \to \infty$ according to the central limit theorem. Three cases are performed for $n = 40$,100 and 250 in this example.

Obtained results are provided in Table 4. Reference values estimated by direct MCS with a number of 1.0×10^6 samples are respectively 1.82×10^{-3} for $n = 40$, 1.73×10^{-3} for $n = 100$ and 1.58×10^{-3} for $n = 250$. The value of ε_1 and ε_1 for the stopping criterion are respectively set to 1.0×10^{-4} and 1.0×10^{-5} and the size of U is 1.0×10^6. The results given by SS-SVM and by ASVR-MCMC are respectively taken from Bourinet et al. [20] and Bourinet [34]. It is seen that all three approaches lead to similar failure probability to the one evaluated by direct MCS; a maximum ε_{Pf} is 7.14% coming from SS-SVM at $n = 40$. Besides, it is found, in all methods, that the required number of calls increases with the number of random variables. However, it is of highly interest to see that the ASVM-MCS significantly outperforms SS-SVM in terms of computational efficiency; for example, in the case of 100 random variables, SS-SVM needs 6036 calls but ASVM-MCS only requires 810 calls for an equivalent accuracy. Whereas, compared with the ASVR-MCMC for which the limit state function evaluation are respectively 616 for $n = 100$ and 1264 for $n = 250$, the ASVM-MCS appears to be less efficient.

The normalised failure probability as a function of the number of calls to the limit state function is plotted in Fig. 10 for the case of 100 random variables in this example. It is found that the failure probability starts to converge at approximately 600 calls to the limit state function.

4.4. Tunnel face stability

The previous three examples can be regarded as academic validation with explicit limit state functions. In this example, a practical geotechnical problem is considered which is about the face stability of a circular tunnel drilling in rock masses. In this context,

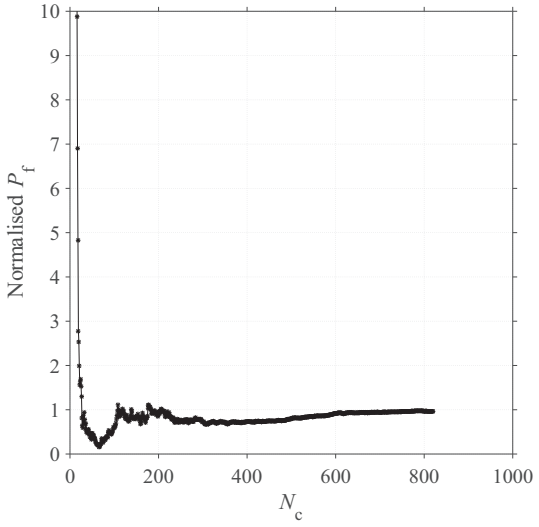


Fig. 10. Evolution of normalised $\widehat{P}_f$ with N_c in Example 3 for $n = 100$.

Table 4
Results of Example 3.

n	Method	N_c	$\widehat{P}_f$	COV($\widehat{P}_f$)	ε_{Pf} (%)
40	Direct MCS	1.0×10^6	1.82×10^{-3}	4.3	–
	ASVM-MCS	341	1.78×10^{-3}	2.4	2.20
	SS-SVM	3729	1.95×10^{-3}	2.8	7.14
100	Direct MCS	1.0×10^6	1.73×10^{-3}	2.4	–
	ASVM-MCS	810	1.72×10^{-3}	2.4	0.58
	ASVR-MCMC	616	1.70×10^{-3}	–	1.73
	SS-SVM	6036	1.74×10^{-3}	2.2	0.58
250	Direct MCS	1.0×10^6	1.58×10^{-3}	2.5	–
	ASVM-MCS	2363	1.57×10^{-3}	2.5	0.63
	ASVR-MCMC	1264	1.56×10^{-3}	–	1.27
	SS-SVM	10,707	1.61×10^{-3}	2.5	1.90

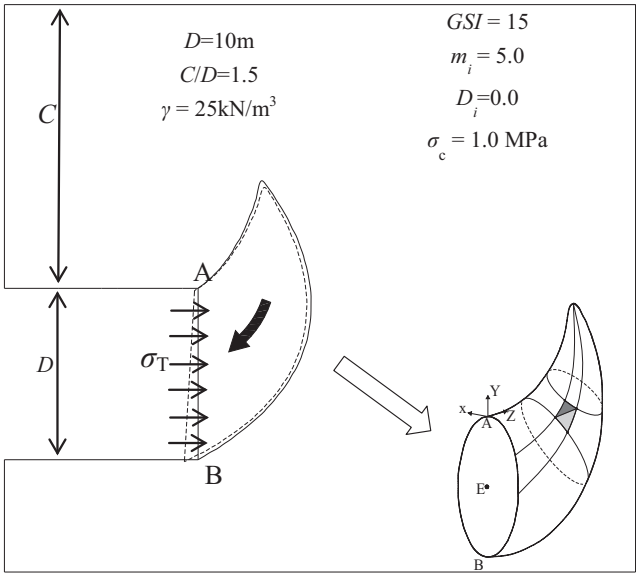


Fig. 11. A sketch of the longitudinal section of a tunnel face.

Table 5
Statistical properties of random variables in Example 4.

Variables	Mean (μ)	COV (%)	Distribution Type
σ_c (MPa)	1.0	22	Normal
GSI	15	18	Normal
m_i	5.0	11	Normal
γ (Kn/m^3)	25	4.7	Normal
σ_T(kPa)	70– 90	15	Normal

an implicit limit state function may be involved. Fig. 11 shows a sketch of the longitudinal section of a tunnel face with a diameter D and a buried depth C. In order to avoid tunnel face collapse during the excavation, it is necessary to provide a support to the tunnel face [36,38,39], idealized as a face pressure σ_T. The generalized Hoek-Brown failure criterion [30,37] is used to characterize the failure of rock masses, which requires four input parameters, the geological strength index (GSI), the uniaxial compressive strength of the intact rock (σ_c), the rock material constant (m_i), and the disturbance coefficient (D_i). In this example, five variables are considered as random variables, as listed in Table 5 for their statistical properties. Other model parameters (tunnel diameter D, the cover depth C and the disturbance coefficient D_i) are taken as deterministic. Thus, a limit state function involving five random variables for the tunnel face stability driven in rock masses is defined as

$$G(\sigma_c, GSI, m_i, \gamma, \sigma_T) = \sigma_T - T(\sigma_c, GSI, m_i, \gamma)$$

where γ is the unit weight of rock masses; T is an implicit function which gives an estimate of the necessary face pressure. An advanced 3D rotational collapse mechanism (see in Fig. 11) developed by Mollon et al. [31] in the context of the kinematic approach of limit analysis is employed herein to predict the necessary face pressure. For a safe excavation, the applied face pressure σ_T should be greater than this minimum necessary face pressure. Reader is referred to Mollon et al. [31] and Senent et al. [32] for more details on T.

Obtained results are given in Table 6 corresponding to mean values of σ_T changing from 70 kPa to 90 kPa. The value of ε_1 and ε_1 for the stopping criterion are respectively set to 5.0×10^{-4} and 1.0×10^{-5} and the size of U is equal to 1.0×10^6. The results provided by direct MCS with 2.0×10^5 samples are taken as reference values. The results provided by the Collocation-based Stochastic Response Surface Method (CSRSM) are also given for comparisons. The Collocation-based Stochastic Response Surface Method aims to approximate the original physical model response using the polynomial chaos expansion. A 5th order polynomial chaos expansion is used to build a response surface.

It is observed in Table 6 that ASVM-MCS provides much better estimates to the failure probability compared to CSRSM in all cases, but requires around 2–3 times less calls to the limit state function. The method ASVM-MCS proposed gives an estimate of failure probability exactly the same as the reference value at $\sigma_T = 70$ kPa (with the maximum cost of 151 calls), and the maximum error is only 0.89% at $\sigma_T = 80$ kPa; however, an error range between 12.39% and 15.21% is observed in CSRSM with 378 calls to the limit state function T. This example proves that ASVM-MCS is able to deal with implicit limit state functions used in practical engineering.

4.5. Discussion

Four applications, respectively involving nonlinearity, high-dimension and implicit limit state function, are used to illustrate the applicability of the proposed ASVM-MCS. The comparison with other methods in literature shows that the proposed ASVM-MCS is a powerful reliability method with respect to efficiency as well as accuracy. Some discussions based on the results obtained are described as follows:

(1) ASVM-MCS shows great performance over SS-SVM with respect to computational efficiency. In the example 1 and example 3, with the same level of accuracy, ASVM-MCS generally needs about $5 \sim 10$ times less call to the limit state function than that of SS-SVM. ASVM-MCS only constructs one SVM classifier based on the active learning strategy, but SS-SVM builds multi-SVM classifiers according to the subset y_i-threshold value. This may be a main reason resulting in huge efficiency gap between two algorithms. Besides,

Table 6
Results of Example 4.

σ_T (kPa)	Method	N_c	$\widehat{P}_f$	COV($\widehat{P}_f$) (%)	ε_{Pf} (%)
70	Direct MCS	2.0×10^5	2.89×10^{-2}	1.4%	–
	ASVM-MCS	151	2.89×10^{-2}	2.2%	0.0
	CSRSM	378	2.53×10^{-2}	0.6%	12.45
80	Direct MCS	2.0×10^5	1.13×10^{-2}	2.2%	–
	ASVM-MCS	120	1.12×10^{-2}	2.2%	0.89
	CSRSM	378	0.99×10^{-2}	1.0%	12.39
90	Direct MCS	2.0×10^5	4.60×10^{-3}	3.4%	–
	ASVM-MCS	115	4.57×10^{-3}	3.4%	0.65
	CSRSM	378	3.90×10^{-3}	1.6%	15.21

ASVM-MCS algorithm is much more easily implementable than SS-SVM. SS-SVM is expected to show good performance in terms of very small failure probability since SS is designed for weak failure probability calculations. Therefore, a possible extension to ASVM-MCS in the future is to incorporate the Importance Sampling or Subset Simulation into ASVM-MCS algorithm for very small failure probability [33–35].

(2) Unlike previously published adaptive SVM [14,23], the learning scheme of the proposed ASVM-MCS does not require a global optimization algorithm. Instead, it actively chooses the next point from the pool U generated at the first step. This makes it easily implementable and highly efficient, especially for high-dimensional problems. The proposed adaptive SVM depends not only on the learning function, but also on the quality of available samples in U. Therefore, the size of U should be large enough to offer informative samples which are most benefit to the updating of a trained SVM. However, if the size of U is too big, the calculation time of the active learning process would be enlarged and more iteration steps would be needed to meet the stopping criterion. Therefore, a compromise should be made between the accuracy and the computational cost. In this work, a U with 1.0×10^6 samples for each example is proved to be a good choice.

(3) Since the computation of failure probability is based on the Monte Carole samples, the proposed ASVM-MCS is not required to make correct classifications over the whole design space, but only in the region at which most Monte Carlo samples are located. This indicates that the limit state function is allowed to be roughly approximated at domain where Monte Carlo samples are rare. Besides, the first three examples addressed in this work are characterized by smooth limit state surfaces. This is the reason that the selected Gaussian kernel performs well for all these examples.

(4) All curves showing normalised $\hat{P}_f$ as a function of N_c reveal that the normalised $\hat{P}_f$ converges before the stopping criterion. Therefore, it is possible to further enhance the efficiency of the proposed ASVM-MCS by using a new stopping criterion that checks the convergence of normalised $\hat{P}_f$.

(5) The proposed ASVM-MCS is only applicable for the assessment of failure probability. It cannot provide other results related to probabilistic analysis, for example, the statistical moments of model response or the sensitivity analysis. This may be a limitation of the proposed ASVM-MCS. But it can easily understand that more information requires higher computational burden. The feature that the SVM only focuses on the classification of model response (the "sign" of model response) is probably a reason that contributes to the high efficiency of the proposed method.

(6) Eq. (6) is well-known for the coefficient of variation calculation of the failure probability by Monte Carlo Simulation. The proposed ASVM-MCS takes advantage of the adaptive SVM to approximate the original limit state surface and then estimates the failure probability by means of MCS. In this context, Eq. (6) is able to give an approximated coefficient of variation estimation of the failure probability computed by the proposed ASVM-MCS. It should be noted that Eq. (6) neglects the error caused by replacing the original limit state function by a SVM. However, the errors of the estimated failure probabilities by the proposed ASVM-MCS are under control and rather small, as it can be seen on the last columns of Tables 1, 3, 4, 6.

5. Conclusion

The probabilistic analysis has gained increased attention in the engineering design recently, as it permits to study the influence of uncertainties of input parameters in a rational way. This paper presented an efficient method named as ASVM-MCS for reliability analysis. This method treats the reliability computation as a classification problem. An active learning strategy is designed to construct a SVM classifier at the minimum number of training samples. A learning function is proposed to sequentially select most informative training samples that both minimize the distance to the SVM decision function and maximize the distance to points of the current DoE for updating the SVM classifier. The enrichment of the DoE is stopped according to the number of samples that are located inside the SVM margin. Then MCS is employed to compute the failure probability based on the SVM classifier obtained. New samples are added to the MC samples in case of a too high coefficient of variation of the failure probability estimated from the metamodel.

The proposed method is validated on three benchmark problems and a practical engineering problem (the stability of a tunnel face driven in rock masses). It is found that the proposed ASVM-MCS is rather economic, requiring low number of evaluations of limit state function with high accuracy. ASVM-MCS is a promising approach for reliability computation, particularly for nonlinear and high-dimensional problems.

A future extension of the proposed ASVM-MCS is to combine the adaptive Support Vector Machine with the Importance Sampling or the Subset Simulation. This extension is expected to develop an efficient approach for assessment of very small failure probability.

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